POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2015
Programme: BE Full Marks: 100
Course: Computer Networks Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is computer network? Write down the merits and demerits of a computer network. 7

   b) Explain the functions of application, presentation and session layers of OSI model in brief. Explain how a bridge updates its bridge table. 8

2. a) Describe how hamming code works for data of 6 bits. Show that it can detect single bit error using an example. 7

   b) Compare circuit switching and packet switching. Draw necessary diagram to illustrate. 8

3. a) A bit stream 10011101 is transmitted using the standard CRC. The generator polynomial is x^3+1. Show the actual bit string transmitted. Suppose the third bit from left is inverted during transmission. Show that this error is detected at receiver end. 8

   b) What do you mean by port and socket? Make comparison between TCP socket and UDP socket. 7

4. a) What is unicast and multicast routing? State and explain distance vector routing with an example. 8

   b) What is a firewall? Explain about virtual private network with diagram. Also illustrate the application of VPN in real world scenario. 7

5. a) P.U. affiliated College is planning to design 8 different department for civil, computer, management and so on. Each department consists of 28 computers with IP address 192.168.218.0/27. Explain the process how will you allocate the IP address and subnet for each of the departments using above IP. 8

   b) What is Virtual Circuit Switching? Explain how Routers build routing table using RIP. 7

ı) Explain the concept of leaky bucket algorithm and explain how does solve the problem of congestion in network?

-) What is DNS server and Queries? Explain any one email serve protocol?

Vrite short notes on: **(Any two)**

- DHCP Principle
- IPSEC
- SMTP

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Network

Semester: Spring

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a)	You are appointed as a communication and network security engineer of new multinational company. How you will choose the new Network topology for your organization? Discuss with a suitable example.	8
b)	Write about bridge and repeater. Differentiate between OSI and TCP/IP. model.	3+4
a)	How optical fiber differ from coaxial cable? How packet switching differs from circuit switching? Explain.	4+4
b)	What are the services of transport layer? Write about TDM and FDM.	3+4
a)	What is Flow Control? How sliding window mechanism is managed through data link layer.	7
b)	Design an algorithm for CSMA/CD with a suitable example.	8
a)	The existing network of Pokhara University (10.200.100.0/22) is to be divided into network of 4 different schools. Among 4 schools two schools need to be subdivided into 3 different departments. Provide a complete IP Address Plan which includes Network Address, Broadcast Address, Usable IP Pool, Subnet Mask and Wildcard Mask.	8
b)	Differentiate between classful and classless routing. Compare & contrast Link-State and Distance-Vector routing algorithm.	3+4
a)	Why congestion occurs in the network? Explain various congestion control algorithm in detail with a suitable example.	7
b)	Discuss the role of DNS. Explain the operation of FTP in corporate Networks.	8
a)	Explain domain addressing. Write about Network management system with its principle components.	3+4

b) What is firewall? Explain its type. Explain the importance of digital signatures.

Write short notes on: (**Any two**)

a) VPN

b) Virtual Circuit Switching

c) Socket Programming

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2016
Programme: BE Full Marks: 100
Course: Computer Networks Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the merits and demerits of computer network. List the merits of client-server model over peer-to-peer model. 7
   b) Define interfaces and services. Explain in brief OSI reference model. 8
2. a) Why flow control is required in modern communication system? Explain with example. Write about Hamming code. 8
   b) Explain ISDN architecture. Differentiate circuit and packet switching. 7
3. a) Write the functions of data link layer. Design an algorithm for CSMA/CD. 7
   b) An organization consists of 3 different departments with 20, 24 and 30 computers. Explain how you will design three subnets for the departments by subnetting 192.168.1.0/24 network. Provide network address, broadcast address, subnet mask, wildcard mask and usable IP pool for each subnet. 8
4. a) What are routing algorithms? Discuss their significance and suitability. Differentiate between classful and classless routing. 8
   b) What are the services of transport layer? Write about TCP and UDP sockets. 7
5. a) Why congestion control is important in communication service networks? Differentiate between Open-loop and Closed-loop congestion control. 7
   b) "HTTPS is preferred over HTTP for Secured Transaction". Justify. Write about different protocols used while sending and retrieving mails. 8
6. a) What is data encryption and decryption? Explain the symmetric key cryptography for ensuring data security. 8

b) What is VPN? Explain the importance of Diffic-Hellman key exchange protocol.

Write short notes on: **(Any two)**

a) RSA

b) CRC

c) Proxy server

POKHARA UNIVERSITY

Level: Bachelor
Semester: Spring
Year : 2016
Programme: BE
Full Marks: 100
Course: Computer Network
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) What is client server network architecture? How it is more secure than peer-to-peer networks? 7

b) What are the reasons for using layered protocols? Explains the functions of various layers in OSI reference model. 8

a) Compare copper wire and fiber optics with respect to the transmission speed, installation/maintenance, security, cost effectiveness, reliability and flexibility. 8

b) What are the factors that cause congestion? Explain Leaky bucket and token bucket traffic shaping. 7

a) How ISDN interfaces and channel works? 7

b) Explain the use of Medium Access Control and Logical Link Control over LAN architecture. 8

a) Explain and draw a typical cell format of an ATM cell. What is the impact of cell payload in ATM network performance? 5

b) What are the principles of using symmetric key and public key Cryptography? Explain any one of public key techniques. 5

c) Differentiate between HTTP and HTTPS protocol. 5

a) Why we need subnet? If the given IP-address of a network is 192.168.10.0/25, then calculate the number of subnets and number of hosts per subnet. 8

b) Discuss about sliding window protocol. Describe mechanism of CSMA/CD along with its application area. 7

a) What is Routing Metrics? Explain different protocols used by Interior Gateway Protocols and Exterior Gateway Protocols of IP-Routing Protocols. 6

b) Why do we need port addressing even though we have local address? Explain about TCP and UDP services and header format. 5

c) Explain the functionalities of switches, router, bridges, and repeaters. 4

Write short notes on: **(Any two)** 2×5

a) VPN
b) DNS
c) CIDR

POKHARA UNIVERSITY

evel: Bachelor | Semester: Fall | Year : 2017
rogramme: BE | | Full Marks: 100
ourse: Computer Networks | | Pass Marks: 45
| | | Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

Explain different topological models in computer network. 8

Discuss the TCP/IP protocol stack with a suitable example. Also Compare TCP/IP with OSI 7

Explain the terms bandwidth, throughput and latency. Explain the working principle of satellite communication system. 7

Compare circuit switching and packet switching. Draw necessary diagram to illustrate. 8

Explain channel access mechanism in CSMA/CD. 8

What is Error Control? Show how Forward Error Control (FEC) technique will help to detect and correct the error 7

Explain HTTP protocol. Describe the difference between SMTP, POP and IMAP servers 7

The existing network of Pokhara University (172.31.255.0/22) is to be divided into network of 4 different schools. Among 5 schools two schools need to be subdivided into 4 different departments. Provide a complete IP Address Plan which includes Network Address, Broadcast Address, Usable IP Pool, Subnet Mask and Wildcard Mask. 8

Explain the difference between distance vector routing and link state routing. 7

What is the role of UDP protocol? Discuss TCP and UDP socket in terms of data transmission and security. 8

What is Virtual Circuit Switching? Explain how Routers build routing table using RIP. 7

Explain public key cryptography. Explain Diffie-Hallman key exchange. 8

rite short notes on: **(Any two)** 2×5

Virtual Private Networks

Email Server Protocl: SMTP,POP, IMAP

TCP socket

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2017
Programme: BE | | Full Marks: 100
Course: Computer Networks | | Pass Marks: 45
| | Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is client active network model? How client server architecture is more secure than peer-to-peer networks? 7

   b) Define Protocol Stack. Compare the different types of networking devices hub, bridge, switch and router. 8

2. a) Compare circuit switching, message switching and packet switching. Explain ISDN signaling and architecture 8

   b) Differentiate between error detection and error correction. A bit string 01101111111001111101111111111100000 needs to be transmitted at the data link layer, what is string actually transmitted after bit stuffing, if flag patterns is 01111110? 3+4

3. a) Discuss the working principle of CSMA. How is it different from CSMA/CD? Explain with necessary diagram. 7

   b) Define class full and classless Routing. Explain the process of RIP and OSPF. 8

4. a) Define BGP? If the given IP-address of a network is 192.168.10.0/27, then calculate the number of subnets and number of hosts per subnet. 8

   b) Define open loop and closed loop congestion control? Explain Leaky bucket and token bucket traffic shaping 7

5. a) Define DHCP. Explain the iterative and recursive DNS query for name resolution with suitable figure. 8

   b) What is SNMP? Explain the advantages of using network management tools. 7

6. a) Define cryptography. Explain the working principle of RSA. 8

   b) What is hand shaking? Explain TCP and UDP services and header format. 7

7. Write short notes on: **(Any two)**

a) IPSEC

b) Frame Relay

c) IPv6

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Year : 2018
Programme: BE
Full Marks: 100
Course: Computer Network
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) You are assigned to design a network infrastructure for your college. Recommend a network solution with hardware and software in current trend that can be used in the college. Make necessary assumption and justify your recommendation with logical argument where possible. 7

   b) Discuss the Seven Layer of OSI protocol stack. Also compare TCP/IP and OSI with a suitable example. 8

2. a) Explain the parameters for analyzing network parameters. How packet switching differs from circuit switching. 8

   b) Differentiate between error detection and error correction. A bit string 0110111111100111110111111111100000 needs to be transmitted at the data link layer, what is string actually transmitted after bit stuffing, if flag patterns is 01111110? 7

3. a) Explain IPv4 header format. Differentiate between IPv4 and IPv6. 8

   b) Design an algorithm for CSMA/CD with a suitable example. 7

4. a) The APNIC Pool for Pokhara University (103.16.32.0/22) is to be divided into network of 7 different schools. Among 7 schools 3 schools need to be subdivided into 2 different departments. Provide a complete IP Address Plan which includes Network Address, 7

Broadcast Address, Usable IP Pool, Subnet Mask and Wildcard Mask.

b) Differentiate TCP and UDP with a suitable example.

5. a) Why congestion occurs in the network? Explain the types of closed loop congestion control mechanism.

b) Define DHCP. Explain the iterative and recursive DNS query for name resolution with suitable figure.

6. a) What is SNMP? Explain the advantages of using network management tools.

b) Explain the working principle of RSA algorithm. If N = 119, public key E=5, and private key D =77, then demonstrate how to send the character plain text F=6 using RSA.

7. Write short notes on: (Any two)

a) Interior Routing Protocol

b) Frame Relay

c) DNS Server

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2018
Programme: BE | Full Marks: 100
Course: Computer Network | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Define Local Area Network. Discuss on any five applications of Computer Network. 7

b) Why TCP/IP is called implementation model? Explain in brief about OSI model. 8

a) What are different guided Medias? Explain each in brief. 7

b) What are the main functions of Data Link Layer? Discuss any two flow control mechanisms. 8

a) Discuss on CRC error detection and Hamming code. 8

b) What are the special IP addresses used in Classful addressing? A multi-national company is granted a site ip172.16.0. 15. Design an IP table with its subnets. 7

a) Define routed and routing protocol. Explain Distance Vector routing algorithm. 8

b) What do you understand by port addressing? Explain TCP header format. 7

a) How does congestion occur in computer network? What are the different mechanisms used in closed loop congestion control? 8

b) Discuss on protocols HTTP and SMTP. 7

a) What is cryptography? Write and explain the RSA algorithm. 8

b) Discuss on Virtual Private Network with its security issues. 7

Write short notes on: **(Any two)** 2×5

a) Peer to Peer Model

b) Frame Relay

c) Proxy Server

POKHARA.UNIVERSITY

Level: Bachelor Semester: Fall Year : 2019
Programme: BE Full Marks: 100
Course: Computer Networks Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) Define Converged Networks. Discuss the merits and demerits of Computer Networks with a suitable example. 7
b) What do you mean by Routing device? Explain design issues of layers. 8
a) Explain twisted pair cable on basis of Categories, Connector used, Performance and Application along with its suitable diagram. 8
b) What are the functions of LLC and MAC sub-layer? Discuss different farming approaches used in data link layer. 7
a) What is Error Correction? Show how FEC technique will help to detect and correct the error with suitable example. 8
b) Explain the frame format of IPV4 with a suitable diagram. 7
a) Differentiate between adaptive and non-adaptive routing. Explain about any intra-AS routing protocol. 8
b) Pokhra University has 3 sub division located at Pokhara as head office, Kathmandu as Examination office and Biratnagar as Contact office with 125, 60 and 29 hosts respectively. Now you as network administrator design the network with below details. 7
i. All the LANs must implemented router as default Gateway.
ii. Ensure that network is secure from inside and outside the network.
iii. Calculate *Broadcast, Network, usable address along with subnet and wild card mask*
iv. ISP provide IP address was 10.0.17.0/24
a) Differentiate TCP and UDP with a suitable example 8
b) What do you mean by congestion and source of congestion? Explain 7

traffic shaping in detail.

6. a) Discuss the role of DHCP. Explain the operation of DNS in corpo[rate] Networks.

   b) What do you mean by Network security? Explain the operatio[n of] Data Encryption Standard Algorithm.

7. Write short notes on: **(Any two)**

   a) Integrated Services Digital Network

   b) UDP-Connection less

   c) Virtual Private Networks

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2019
Programme: BE | Full Marks: 100
Course: Computer Network | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Computer Network. Briefly explain the different types of network model. 7

   b) Why do you need layered Architecture in network? Compare OSI model with TCP/IP model. 8

2. a) Define transmission media. Explain different types of transmission media in detail. 7

   b) What do you mean by framing? List the different framing technique and illustrate bit stuffing with an examples. 8

3. a) Explain the working principle of selective repeat ARQ and point out the merit and demerit of it over Go-Back –NARQ. 7

   b) You are given IP Address 150.152.0.0. you need to subnet the given IP into five different Departments. Perform the subnetting and find the subnet mask, Network Address, Broadcast address and usable host address in all subnet. 8

4. a) Explain IPV4 Header format? Differentiate between IPV4 and IPV6. 7

   b) What is socket address and communication? Explain the services provided by the transport layer. 8

5. a) What is congestion? Briefly explain different types of technique for Traffic shaping. 7

   b) What is SSL? Explain how a request initiated by a HTTP client is served by a HTTP srever. 8

6. a) What is network security? How can firewalls enhance network security? Explain how firewalls can protect a system. 7

   b) Compare symmetric key encryption method with asymmetric key encryption. Encrypt the message "READ" using RSA algorithm. 8

7. Write short notes on: (**Any two**).

   a) Circuit switching and packet switching

   b) Email server protocol: SMTP

   c) VPN

POKHARA UNIVERSITY

Level: Bachelor	Semester:Fall	Year : 2020
Programme: BE		Full Marks: 100
Course: Computer Networks		Pass Marks: 45
		Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

. a) Describe the demerits of computer network. Differentiate between client/server and peer-to-peer network models. 7

b) How are interfaces, protocols and services of layered system related? Discuss the differences between OSI and TCP/IP models. 8

a) List the major function of Internetworking device. Briefly explain the design issues of network Layer? 7

b) What are the metrices of network performances? Describe them in short. 8

a) Explain how selective repeat works with figure. 7

b) Describe framing by character count. How does byte stuffing overcome the disadvantages of character count algorithm? 8

a) Why summarization is important? P.U. is planning to design new network for four different departments. Each department consists of 24 hosts with IP address 172.16.1.0/27. Explain the process how you will allocate the IP address, subnet mask for each of the department using above IP address. 8

b) Explain IPV4 header protocol format and also differentiate between IPV4and IPV6 features. 7

a) Describe 4 way TCP handshaking with suitable diagram. 8

b) Why congestion is prominent in Networks?. Explain Explicit and Implicit control algorithm with a suitable example. 7

a) What is SMTP? Differentiate POP and IMAP protocols. 8

b) What do you mean by Network security? Explain the operation of Data Encryption Standard Algorithm. 7

7. Write short notes on: **(Any two)**

a) CSMA/CD

b) DHCP

c) Firewall

Pokhara University

:l: Bachelor | Semester-Spring | Year: 2020
ramme: BE | Full Marks: 70
rse: Computer Networks | Pass Marks: 31.5
Time: 2 hrs.

ndidates are required to give their answers in their own words as far as practicable.
' figure in the margin indicates full marks.

empt all the questions.

Section A: (5 ×10 = 50)

. No 1 Define Computer Networks. Discuss its merits and demerits and also the area of application in today's world. 10

. No 2 What do you understand by Client server and Peer to Peer architecture? Compare and Contrast between them with suitable diagram. 10

OR

Explain about the operation of CSMA/CD technique with its flowchart. Discuss about Go Back N ARQ and Selective Repeat ARQ in brief 10

. No 3 Show how Hamming code technique will be used to compute error detection and correction with an example.(Assume a 7-bit number) 10

. No 4 Compare IPv4 and IPv6 header with their header diagrams. 10

. No 5 Define congestion and explain about leaky bucket and token bucket algorithm in brief. 10

Section B: (1 × 20 = 20)

Q.N 6 a) The existing network of Pokhara university (172.16.0.0/16) is to be subdivided into four different schools located at different states of the country connected with Network Service Provider (NSP). The location details of each school are enlisted below. 10

i. School of Engineering – Pokhara
ii. School of Law – Kathmandu
iii. School of Management – Chitwan
iv. School of Environment Research and Development – Nepalgunj

Design the network with complete IP Address plan for each school.

b) In RSA cryptosystem, a participant A uses two prime numbers $p = 13$ and $q = 17$ to generate her public and private keys. If the public key of A is 35, then what is the private key of A? 10

POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2021
Programme: BE | Full Marks: 100
Course: Computer Network | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Intranet. Which network model is used for connecting devices within Office and why? Explain with neat diagram along with advantages and disadvantages. 8

   b) Define protocols and standards. Compare TCP/IP and OSI reference model. 7

2. a) Why do we use fiber optics for long distance communication? Explain fiber optics single mode of propagation? Describe about network performance bandwidth and latency. Write a command to check latency to server with IP 4.4.8.8 from your computer with Windows/Linux OS. 8

   b) Differentiate between Distance Vector Routing algorithm and Link State Routing algorithm with example. What are features of IPV6 protocol and provide one example of IPV6. 7

3. a) Define codeword. Explain with example how transmission error is detected and corrected using Hamming code. 7

   b) A company have 3 different departments with 65, 32 and 12 network devices. Explain how you will design network for this company from provided network of 10.10.100.0/24. Provide network address, broadcast address, subnet mask, wildcard mask and usable IP pool for each subnet. 8

4. a) Draw IEEE 802.3 Frame format. Explain random access protocol: ALOHA. 8

   b) What do you mean by Socket Programming? Explain TCP Client/Server Socket flow with suitable diagram. 7

5. a) In your opinion what are the main causes of congestion in network. Explain about closed-loop congestion control. 7

   b) Explain about DNS, its importance, name resolution iterated and recursive query with neat diagram. 8

a) What do you mean by symmetric and asymmetric key algorithm Explain RSA with suitable algorithm to perform Key generation fo public key, private key, encryption and decryption.

b) Differentiate between Switch and Hub. Explain exterior routing protocol.

Write short notes on: (**Any two**)

a) Firewall

b) DHCP

c) Proxy Server

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2021
Programme: BE Full Marks: 100
Course: Computer Network Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a)	Explain different topological model in computer network?	8
b)	What do you mean by network performance? Compare between TCP/IP and OSI.	7
a)	What is client active network model? How client server architecture is more secure than peer to peer network?	8
b)	What are the functions of LLC and MAC sublayer? Explain about framing in detail.	7
a)	Determine which bit is in error in the odd parity. Hamming code character is 1100111	8
b)	Illustrate the guided and unguided transmission media for the data communication? How Data flows on the physical layer?	7
a)	Explain IPV4 header format. Differentiate between IPV4 and IPV6	8
b)	What is adaptive and Non-Adaptive Routing? Compare classful and classless routing.	7
a)	What is socket programming? Differentiate between TCP and UDP with suitable diagram.	8
b)	Why congestion occurs in network? Explain leaky Bucket algorithm with suitable diagram	7
a)	Define DHCP, SMTP, POP, and IMAP with their port number?	8
b)	Define types of encryptions used in security. Justify how firewalls can enhance network security.	7

7. Write short notes on: (**Any two**)

a) Hub and Switch

b) Frame Relay

c) VPN

POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2022
Programme: BE | Full Marks: 100
Course: Computer Networks | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Define Computer Network .What are the merits and demerits of using Computer Network? Explain. 8

b) What do you understand by interfaces, protocol and services? Illustrate with detailed diagram the different layers of OSI model. 7

a) Define Transmission Media. Explain about different propagation techniques used in wireless transmission. 8

b). Compare **CSMA/CD** and **CSMA/CA** with the help of flow chart. 7

a). What are the function of LLC and MAC sub-Layer? Why byte stuffing is done? 7

b) Explain TCP and UDP Socket Programming process? Why socket programming is necessary? 8

a) Is it necessary to implement the IPV6 in existing network? Differentiate between IPV4 and IPV6? 8

b) What are the benefits of subnetting. Perform the subnetting of 10.0.0.0 using CIDR value 11. 7

a) How can we control congestion in the network? Explain different types of congestion control techniques. 8

b) Explain Client/Server paradigm. Explain the role of http, static and dynamic document in transporting information. 7

a) What is cryptography? Explain any key exchange protocol in detail. 8

b) What is the role of firewall in network? List out and explain types of firewall. 7

Write short notes on: **(Any two)** 2×5

a) Router and Switch

b) Shortest path routing algorithm

c) Client Server and Peer to Peer Network

POKHARA UNIVERSITY

Level: Bachelor	Semester: Spring	Year : 2023
Programme: BE		Full Marks: 100
Course: Computer Networks		Pass Marks: 45
		Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1.	a)	Discuss the merits and demerits of computer network. Compare star, bus, ring and mesh topology.	4+3
	b)	Explain the working principle of different types of network devices: hub, switch, bridge, and router.	8
2.	a)	Explain circuit switching mechanism. Explain the terms throughput, latency, bandwidth and jitter.	3+4
	b)	What is error control. Explain CRC method of error detection with a suitable example.	2+6
3.	a)	Explain Gigabit Ethernet with help of below parameters: Mac-sub layer, full duplex mode, half duplex mode, topology, implementation and encoding.	7
	b)	What are the draw backs in IPV4? Which of the draw backs do IPV6 solve? Explain.	4+4
4.	a)	Perform the sub netting of 192.168.0.0 using CIDR value 26. What is VLSM and why do we use it?	5+3
	b)	Discuss about Interior and Exterior Routing protocols. What is count to infinity problem in distance vector.	7
5.	a)	What is congestion in a network. Explain about congestion control mechanism.	8
	b)	List out and explain TCP client/server socket flow with suitable diagram.	7
6.	a)	Explain the roles of proxy server. Compare HTTP with HTTPS.	7
	b)	What do you mean by network security? Explain RSA algorithm.	8
7.		Write short notes on: (Any two)	2×5
	a)	Kerberos	
	b)	Optical fiber	
	c)	CSMA/CD	

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Network

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define Computer Network along with its architecture. Explain different kind of services that are benefited by the networking technology in today's world? 4+4

b) Define protocol. Why do we need layered protocol architecture? Discuss each layer of TCP/IP protocol architecture in detail. 7

a) Justify the implications of Optical fiber in present day to enable global communications with a suitable example. 8

b) What is Error Detection? Show how CRC technique will help to detect the error with suitable example. 7

a) Design an algorithm for CSMA/CD with a suitable example. 8

b) Explain the frame format of IPv4 with a suitable diagram. 7

a) *Pokhara University is a Member of APNIC.* It has acquired an IPv4 pool of (103.32.0.0/22). From the acquired pool, it is required to be subnetted and distributed across Six Schools. Among Six Schools, Two Schools need to be subdivided into two different departments. Design and provide a complete IP Address plan which includes Network Address, Broadcast Address, Usable IP Pool, Subnet Mask and Wildcard Mask for every subnet. 8

b) What is TCP? Explain TCP connection and termination with suitable diagram along with its TCP state while connection and termination. 7

a) What is network congestion? Explain how TCP handle congestion. 7

b) Compare Http, Https, document, Html, URL, port in web request. 8

6. a) What do you mean by Asymmetric Cryptography? Write simple out line algorithm in Asymmetric cryptography to generate key, Encryption and Decryption process.

   b) What is packet Filtering firewall? Show how Differ-Hellman works with help suitable example.

7. Write short notes on: **(Any two)**

   a) Interior and exterior routing

   b) Simple Mail Transfer Protocol

   c) Framing technique

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024
Programme: BE | Full Marks: 100
Course: Computer Networks | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the merit and demerits of computer networks. List the merits of client-server model over peer to peer model. 7
   b) Define interfaces and services. Explain in brief OSI reference model. 8
2. a) What is Transmission media. How ISDN interfaces and channel works? 7
   b) Suppose you are a private consultant hired by a company to setup the network for their enterprise and you are given a large number of consecutive IP address starting at 120.89.96.0/19. Suppose that four departments A, B, C and D request 100, 500, 800 and 400 addresses respectively, how the subnetting can be performed so that address wastage will be minimum. 8
3. a) Explain **Gigabit Ethernet** with help of below parameters: Mac-sub layer, full duplex mode, half duplex mode, topology, implementation and encoding. 7
   b) Compare Stop-wait-ARQ, Go-Back-N and Selective Repeat based on below points 8
      a. Mode of communication.
      b. Addressing channel error
      c. Handling error control and buffering
      d. Technique for channel 6utilization
4. a) What is classful and classless address? Differentiate between link state and distance vector routing protocol. 7
   b) What are the functions of the Transport Layer? Explain TCP and its header segment. 8
5. a) What is congestion in a Network? Compare between leaky bucket and token bucket algorithm with the operation how token bucket works 7
   b) What is the role of URLs and Web Browsers in WWW. Explain about DHCP server in details. 8
6. a) What do you mean by Asymmetric key algorithm? Explain RSA algorithm Encryption and decryption process with suitable example 8

b) What is NMS. Explain about major components of Network Management Architecture.

7. Write short notes on: **(Any two)** 2x

a) HTTP
b) IPv4 vs IPv6
c) ATM